Cubic graphs on at most 40 vertices satisfy the Erdős–Gyárfás conjecture: a calibrated SAT-modulo-symmetries search

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Abstract

The Erdős–Gyárfás conjecture states that every graph with minimum degree at least 3 contains a simple cycle whose length is a power of two. Markström (2004) verified the conjecture for cubic graphs on at most 28 vertices, and that exhaustive bound stood until 2026. We report an exhaustive verification for connected cubic graphs on 30, 32, 34, 36, 38 and 40 vertices: in each order there is no cubic graph avoiding cycles of length 4, 8 and 16. Consequently a smallest cubic counterexample, if one exists, has at least 42 vertices. The search uses SAT modulo symmetries with a complete forbidden-subgraph propagator; unlike a decision procedure taken on trust, the instrument is first *calibrated* against exhaustive enumeration: it is required to reproduce, exactly and as sets, thirteen reference censuses of cubic graphs, nine of which have nonzero answers, including a 348.7 core-hour enumeration of order 30 carried out independently by the author. Positive controls are additionally run at the frontier orders themselves. All code, logs and censuses are available for independent verification.

1 Introduction

Erdős and Gyárfás asked whether every finite graph with minimum degree at least 3 contains a cycle of length 2^k for some $k \geq 2$; the problem appears in Erdős's problem papers from 1993 onwards [1], and both believed the answer to be negative. Erdős offered \$100 for a proof and \$50 for a counterexample [3]; the problem is listed with a \$1000 prize as #64 of the

Erdős problems collection [2]. Liu and Montgomery [4] settled the case of sufficiently large minimum degree, and Carr [5] showed that any minimal counterexample is predominantly cubic, so the cubic case is the structurally critical one.

Computational state of the art. Markström [3] generated all cubic graphs on fewer than 29 vertices and found no counterexample; his Table 3 records the numbers of cubic graphs with neither a 4-cycle nor an 8-cycle for $n = 24, 26, 28$, namely 4, 23 and 251. In 2026 Balaji [6] reported, via SAT modulo symmetries, that no graph of minimum degree at least 3 on at most 31 vertices avoids all power-of-two cycles, which yields the bound $n \geq 32$ for the general class and, a fortiori, for the cubic one; that work is a preprint under review, and its documentation states that no machine-checkable certificate is currently available for the claim, since the clauses contributed by the subgraph propagator are not RUP/RAT-derivable from the encoding, and that third-party reproduction is outstanding. Tranquilli [7] treats the cubic *bipartite* subclass and reports the bound $n \geq 60$ there.

Result.

Theorem 1. *Every connected cubic graph on $n \leq 40$ vertices contains a cycle of length 4, 8 or 16. In particular, a smallest cubic counterexample to the Erdős–Gyárfás conjecture has at least 42 vertices.*

Cubic graphs have even order, and a counterexample may be assumed connected, since every component of a cubic graph is itself cubic. Orders $n \leq 28$ are due to Markström; orders 30, 32, 34, 36, 38 and 40 are verified here.

Why forbidding $\{4, 8, 16\}$ suffices. For $n \leq 31$ the admissible power-of-two cycle lengths are exactly $\{4, 8, 16\}$, so avoiding them is equivalent to being a counterexample. From $n = 32$ on, C_{32} also fits, and adding a 32-cycle to the forbidden set inside the search is expensive: at $n = 32$ it is a Hamiltonian cycle, and as a propagator it would be evaluated on every partial assignment. We therefore *decouple*: the search forbids only $\{4, 8, 16\}$, which is a *weaker* constraint, and the larger powers of two are tested afterwards on whatever survives. Since the search returns no graph at all in every order considered, the posterior test is vacuous and the conclusion is stronger than required: not merely "no counterexample", but "every cubic graph of that order already contains one of C_4, C_8, C_{16}".

2 Method

We use SAT modulo symmetries (SMS, Kirchweger and Szeider [8]), which performs complete, isomorph-free graph generation inside a CDCL solver, built together with the Glasgow subgraph solver [9] as a complete forbidden-subgraph propagator. One call per order decides: *is there a connected cubic graph on n vertices containing no C_4, C_8 or C_{16} as a subgraph?* The graph is encoded by the $\binom{n}{2}$ edge variables; cubicity is imposed as a pair of cardinality constraints (sequential counter) fixing every degree to exactly 3, and connectivity is imposed by SMS.

Two configuration choices deserve mention. First, the minimality (canonicity) check of SMS is applied without cutoff; the default cutoff of $2 \cdot 10^5$ truncates the check and thereby weakens complete symmetry breaking. Second, a run that exhausts its time budget is recorded as such and never as unsatisfiability, and a run whose solution count cannot be parsed is recorded as an error rather than as zero; exit codes are checked in every case.

2.1 Calibration of the instrument

An unsatisfiability answer is an assertion of the solver, not an object one can inspect; enumeration, by contrast, produces countable objects that can be compared with published censuses. The dangerous failure mode of a SAT-plus-propagator pipeline is over-constraint: such a pipeline returns zero everywhere and therefore passes every test whose correct answer is zero. Only a test with a known *nonzero* answer detects it.

We therefore require the pipeline to reproduce thirteen reference censuses of connected cubic graphs, nine of them nonzero (Table 1). Three sources of ground truth are used: Markström's published Table 3; an exhaustive enumeration at orders 24–30 performed by the author with `geng` (nauty [10]) and an embedded C pruning hook, 348.7 core-hours at order 30; and, to exercise the C_{16} propagator against nonzero answers, fresh counts of cubic graphs containing no 16-cycle at orders 14–22. At orders 14–20 these counts were computed twice with independent cycle detectors (a bitmask depth-first search and a networkx-based one). Order 22, the largest, was counted once, with the bitmask detector, and guarded differently: it was generated in six `res/mod` shards whose graph totals sum to 7 319 447, the known number of connected cubic graphs on 22 vertices, so the partition neither lost nor duplicated a graph; of those, 52 781 contain no 16-cycle, and each of those 52 781 was re-tested individually with the networkx detector. The agreement of SMS at this order is not counted as verification of the anchor, since SMS is the instrument under calibration.

Order	Forbidden	Expected	SMS
$n = 10$	C_{16}	19	19
$n = 14$	C_{16}	509 (no 16-cycle fits)	509
$n = 16$	C_{16}	219	219
$n = 18$	C_{16}	1471	1471
$n = 20$	C_{16}	12709	12709
$n = 22$	C_{16}	52781	52781
$n = 24$	C_4, C_8	4 (Markström)	4, same graphs
$n = 26$	C_4, C_8	23 (Markström)	23, same graphs
$n = 28$	C_4, C_8	251 (Markström)	251
$n = 24, 26, 28, 30$	C_4, C_8, C_{16}	0	0

Table 1: Calibration against exhaustive enumeration. At $n = 24$ and $n = 26$ the comparison is at the level of sets, not merely cardinalities: the graphs emitted by SMS and by `geng` coincide after canonical labelling with `labelg`.

Two robustness axes were varied. Re-deciding with the totalizer cardinality encoding, a structurally different CNF for the same degree constraint, reproduces 251 at $n = 28$, 1471 at $n = 18$ and zero at $n = 30$, $n = 32$, $n = 36$ and $n = 40$; the last two are frontier orders, re-decided in 4143 s and 17 207 s against 3366 s and 18 922 s for the sequential counter, so the largest order reported here has been decided twice with structurally different formulas. The comparison of graph sets at $n = 24$ and $n = 26$ is exact.

2.2 Positive controls at the frontier orders

Calibration establishes that the encoding and the three propagators are correct, and encoding faults are order-independent: the same code runs at every order. It does not, however, exclude order-dependent breakage that would render the formula vacuously unsatisfiable above some size. We therefore also run existence queries at the frontier orders themselves, forbidding only $\{C_4, C_8\}$, where solutions are known to exist: SMS returns a satisfying graph at $n = 32$, 34, 36, 38, 40, 42 and 44, the last two beyond the orders decided here. The zero counts reported below are thus not an artifact of the pipeline ceasing to find anything at large orders.

n	connected cubic graphs with no C_4, C_8, C_{16}	single-core time
30	0	187 s
32	0	505 s
34	0	1188 s
36	0	3366 s
38	0	8157 s
40	0	18922 s

Table 2: The frontier sweep. Order 30 also serves as calibration: the same answer was obtained by exhaustive enumeration in 348.7 core-hours.

2.3 Results

The observed growth is a factor of roughly 2.3 to 2.8 per two vertices (order 40 came in at 2.32 times order 38), so the method remains within reach of a single desktop machine for several further orders. Order 40 was decided in a single-threaded run of 5 h 15 min; the native cube-and-conquer parallelisation of SMS was measured on this instance family and discarded, as it yields only a factor of 1.7 on twelve workers, and CaDiCaL's formula simplification, which is faster still, does not preserve the set of models: under it the $n = 28$ census returns 241 instead of 251.

3 What is not claimed

The result above order 30 rests on a single method. No machine-checkable proof certificate accompanies it: SMS can emit an LRAT proof, but the clauses contributed by the subgraph propagator are not RUP/RAT-derivable from the encoding alone, so a generic checker cannot validate it; a certified-SMS treatment in the sense of [8] is the natural strengthening and is not attempted here. What we do claim is a calibrated instrument: the pipeline reproduces every reference census available in this class, including eight with nonzero answers and two at the level of graph sets, and demonstrates satisfiable behaviour live at the frontier orders. Order-dependent failure modes are guarded procedurally, not proved absent. Independent reproduction remains desirable, and the artifacts below are published to that end.

4 Data availability

Repository: https://github.com/christopher-manzanovimos_geci/erdos-gyarfas-cubic-front archived deposit: Zenodo, DOI 10.5281/zenodo.21976755. The repository contains the SMS driver, the calibration battery, the ground-truth generators and cycle detectors, the run logs with exit codes and timings, the graph6 censuses for $n = 14$–22 (no C_{16}) and $n = 24$, 26 (no C_4/C_8), the complete July 2026 enumeration apparatus used as independent ground truth, and the build record. Versions used: SMS at commit `464f12f`, CaDiCaL at `b023aaf`, Glasgow subgraph solver at `1217f5b`; note that the SMS build script clones the Glasgow solver at its current head rather than at a pinned commit, so this component should be recorded rather than assumed by anyone reproducing the computation.

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